Sekwencjonowanie następnej generacji

Fundusz Zdolni

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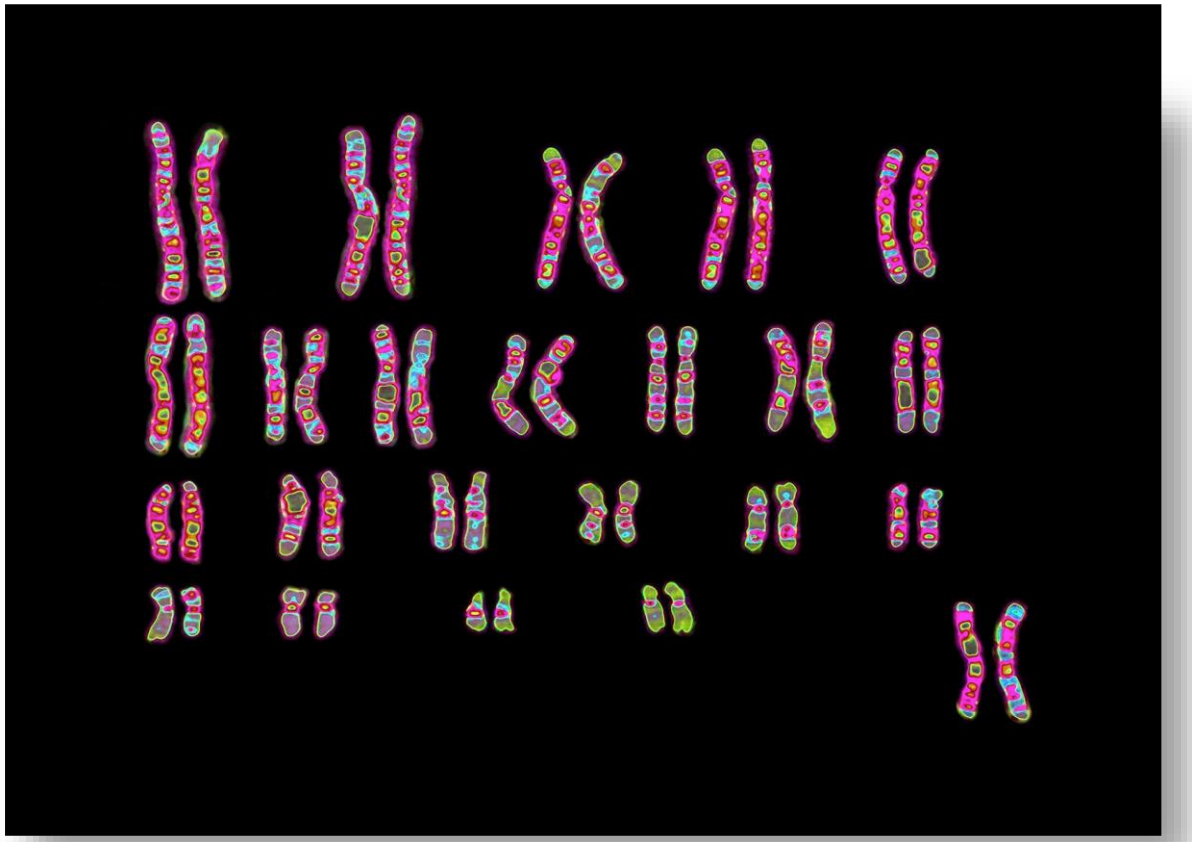
Genomika



Genomika

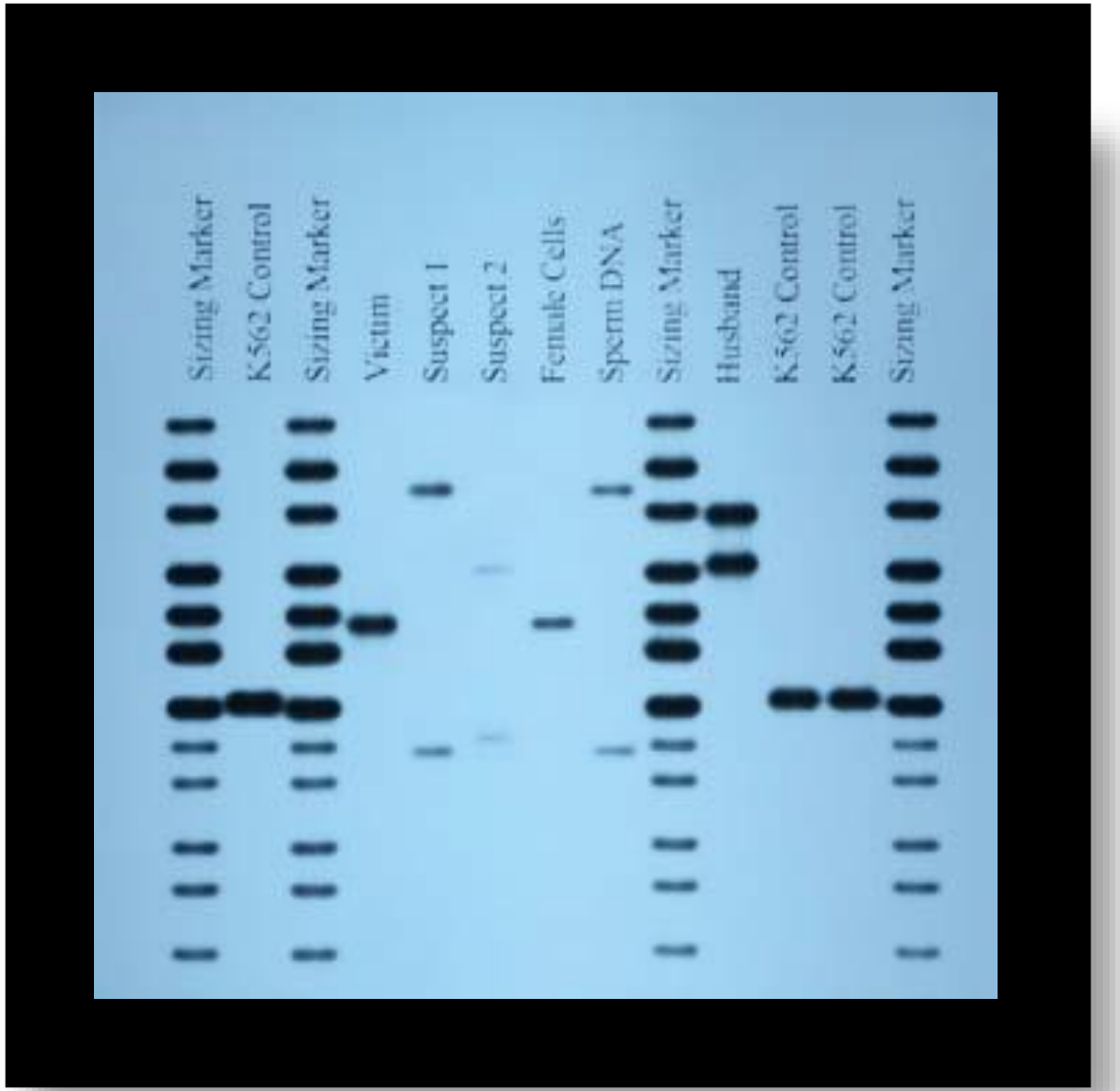
Variable **N**umber **T**andem **R**epeat:

Allel 1:

GGTAGTCGGTAAGAGTCGGTAAGAGTCGGTA
AGAGTCGGTAAGAGTCGGTAAGAGTCGGTAA
GAGTCGGTAAGCGGTAAGTAGCGA

Allel 2:

GGTAGTCGGTAAGAGTCGGTAAGAGTCGGTA
AGAGTCGGTAAGAGTCGGTAAGAGTCGGTAA
GAGTCGGTAAGAGTCGGTAAGAGTCGGTAAG
AGTCGGTAAGCGGTAAGTAGCGA



Reakcja PCR

Denaturacja

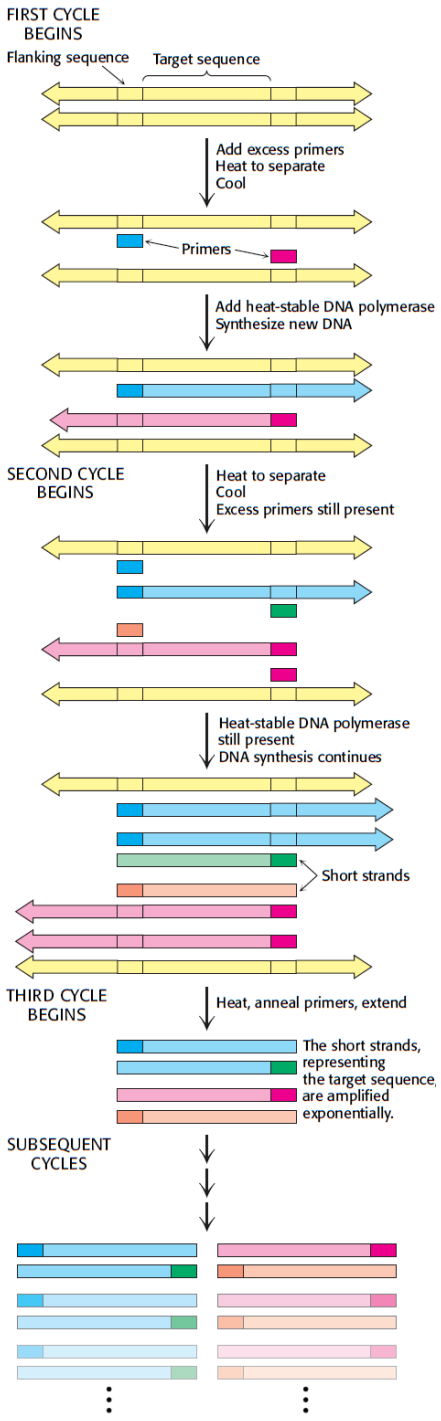
- 95°C

Annealing

- 55°C

Propagacja

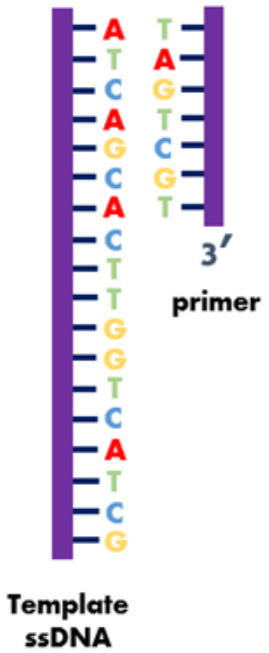
- 72°C

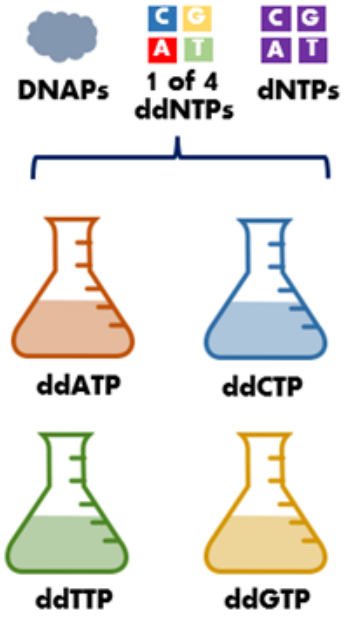


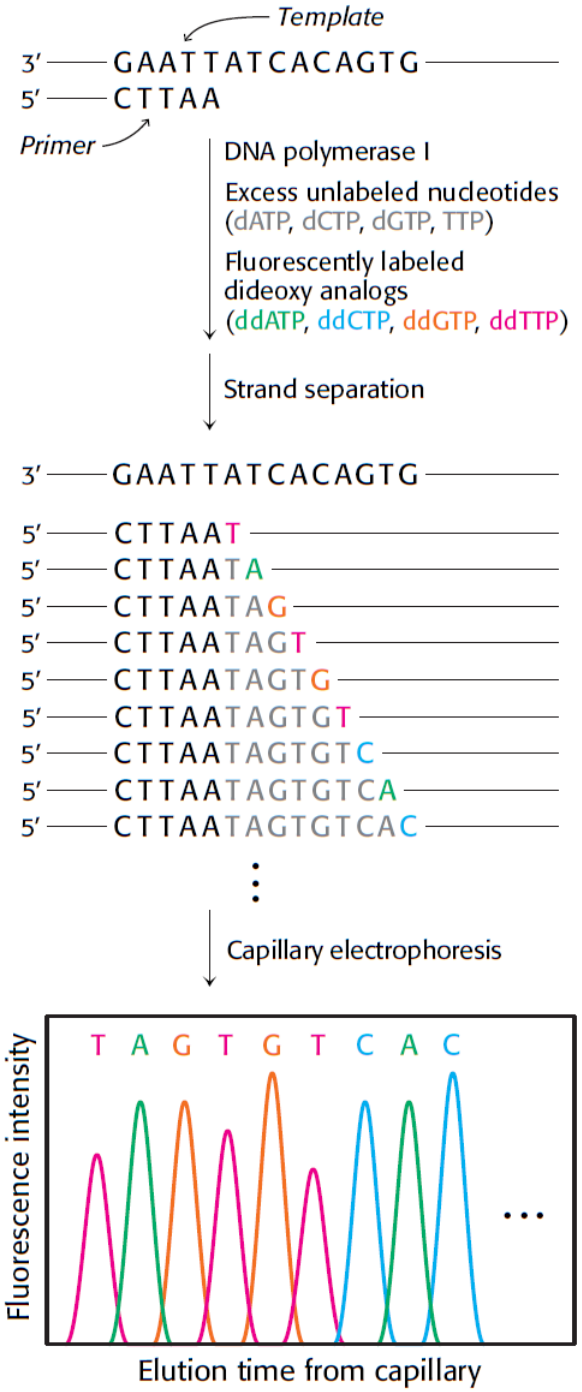
Generacje w sekwencjonowaniu

First generation	Second generation (next generation sequencing)	Third generation
Sanger sequencing Maxam and Gilbert Sanger chain termination	454, Solexa, Ion Torrent, Illumina	PacBio Oxford Nanopore
Infer nucleotide identity using dNTPs, then visualize with electrophoresis	High throughput from the parallelization of sequencing reactions	Sequence native DNA in real time with single-molecule resolution
500–1,000 bp fragments	~50–500 bp fragments	Tens of kb fragments, on average
Short-read sequencing		Long-read sequencing

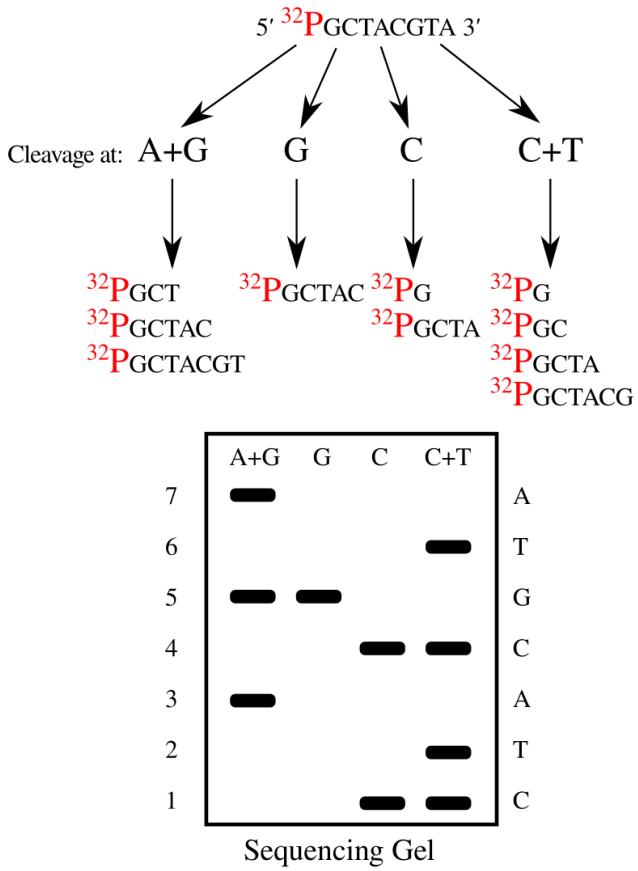
Sekwencjonowanie m. Sangera







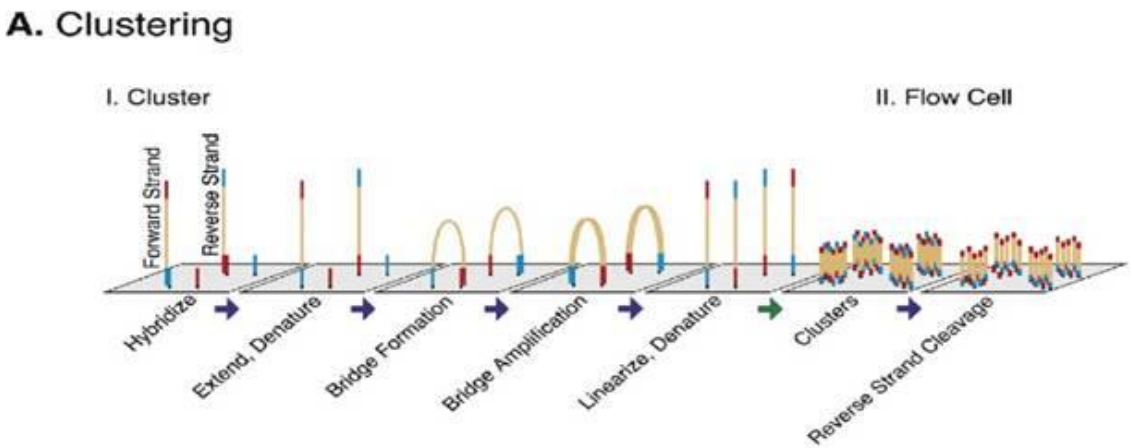
Sekwencjonowanie m. Maxama Gilberta



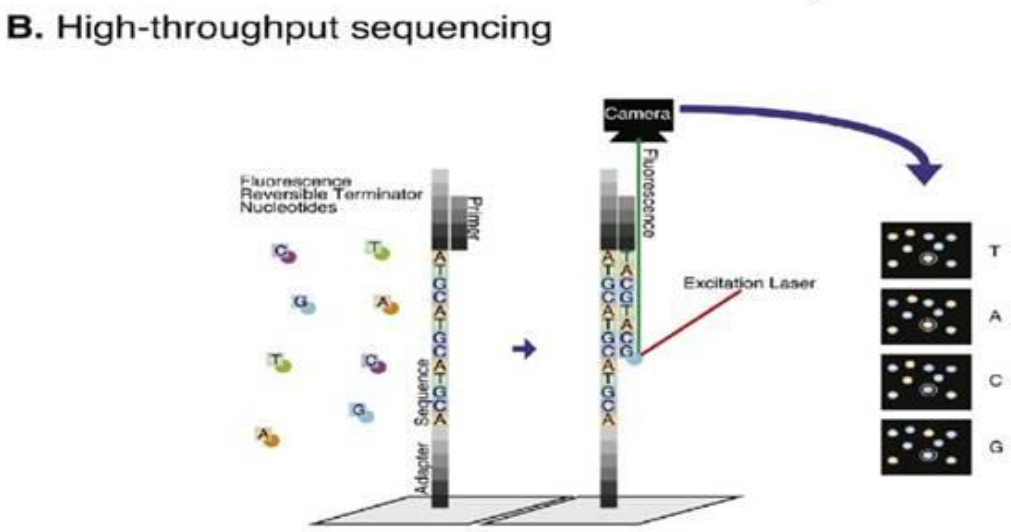
NN	Cleavage specificity	Modifying agent	Modified base displacement conditions	Cleavage conditions
1	G	Dimethyl sulfate	Piperidine	Piperidine
2	G+A	Acid	Acid	Piperidine
3	C+T	Hydrazine	Piperidine	Piperidine
4	C	Hydrazine + NaCl	Piperidine	Piperidine
5	G>A	Dimethyl sulfate	Heating at pH 7	NaOH or piperidine
6	A>G	Dimethyl sulfate	Acid	NaOH
7	A>C	NaOH	Piperidine	Piperidine
8	G	Methylene blue	Piperidine	Piperidine
9	T	OsO_4 or $KMnO_4$	Piperidine	Piperidine
10	C	$NH_2OH \cdot HCl$ pH 6	Piperidine	Piperidine
11	A	Diethyl pyrocarbonate	Piperidine	Piperidine

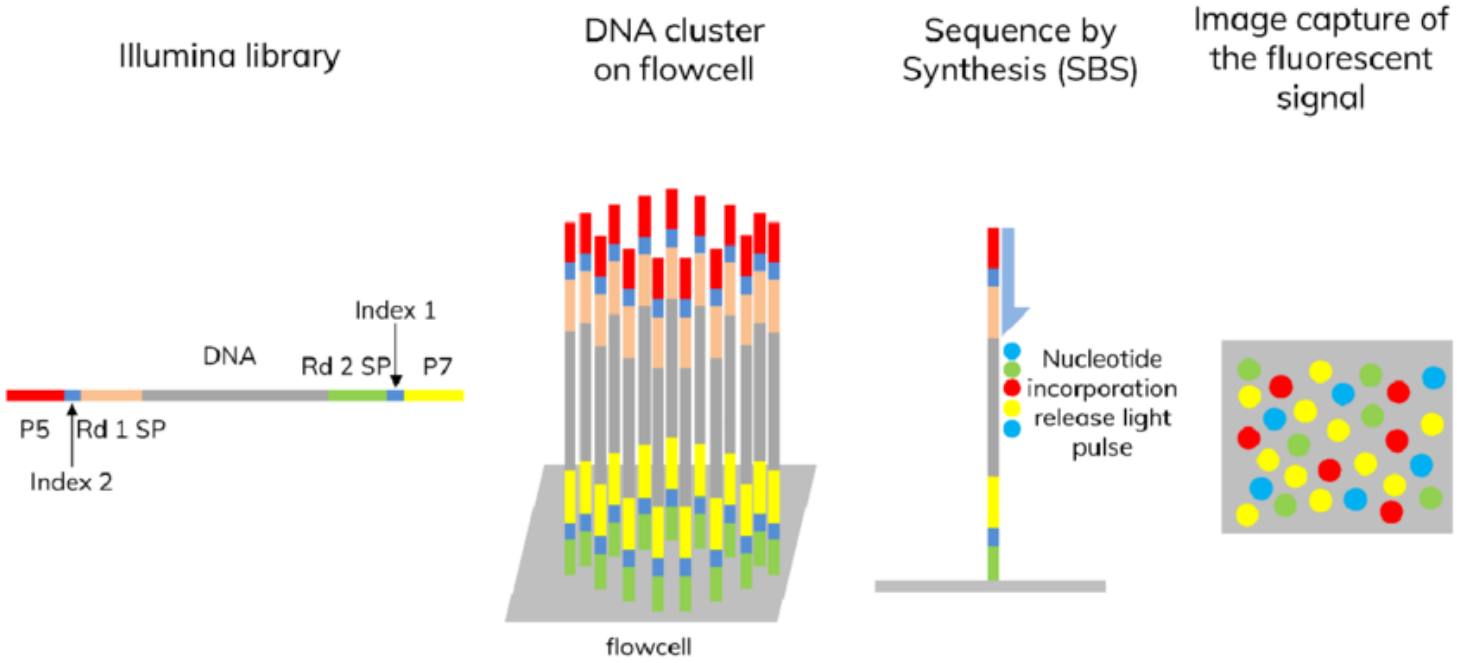
Advanced Organic Chemistry of Nucleic Acids

Sekwencjonowanie przez syntezę

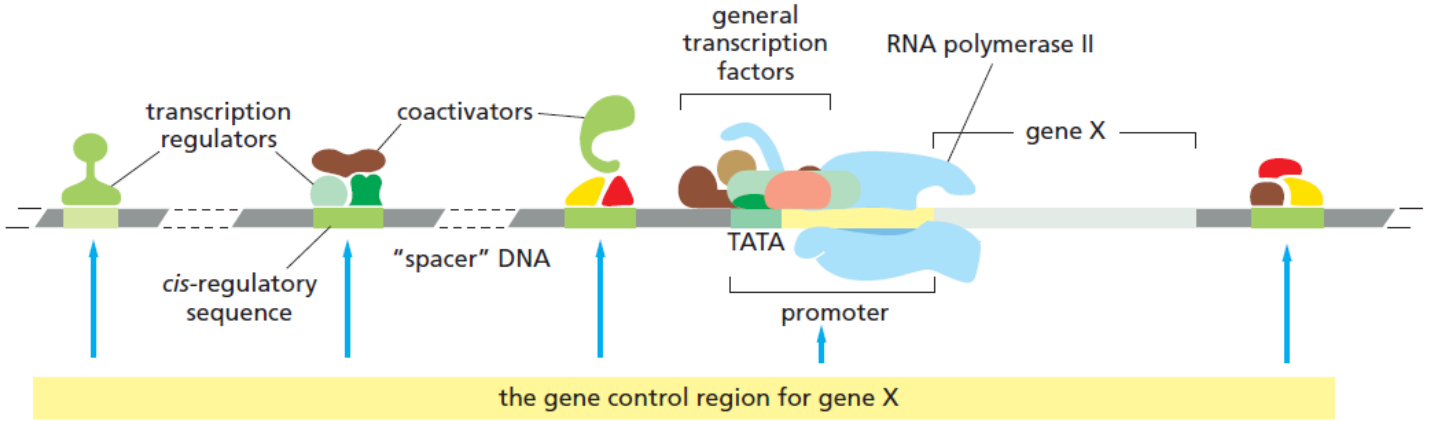


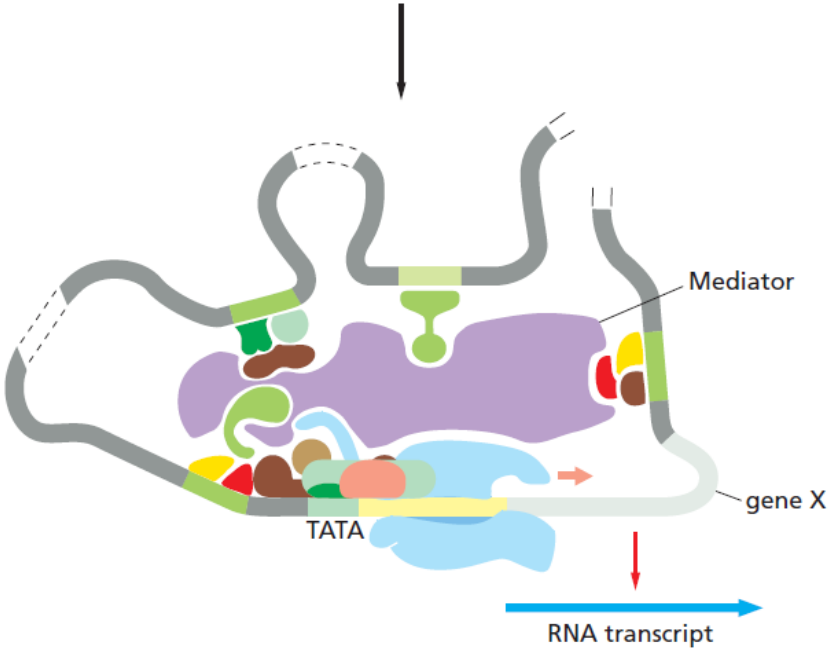
https://www.youtube.com/watch?v=fCd6B5HRaZ8

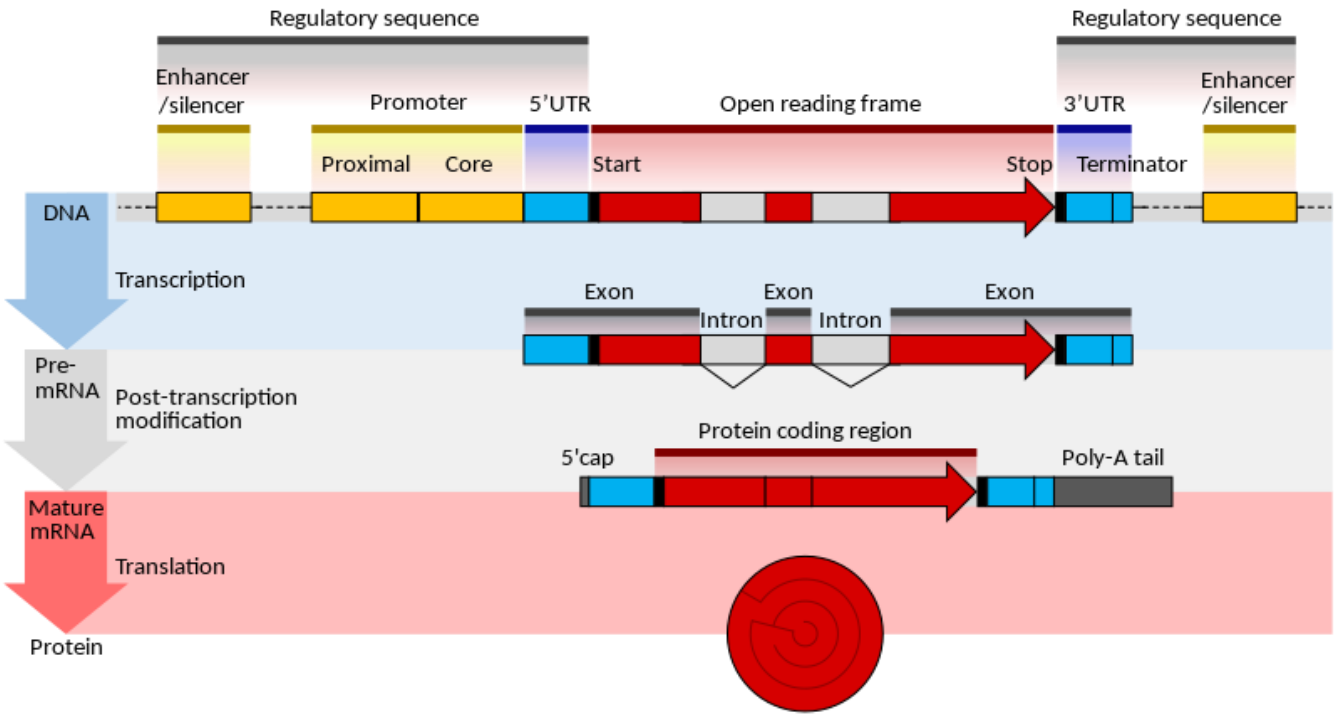




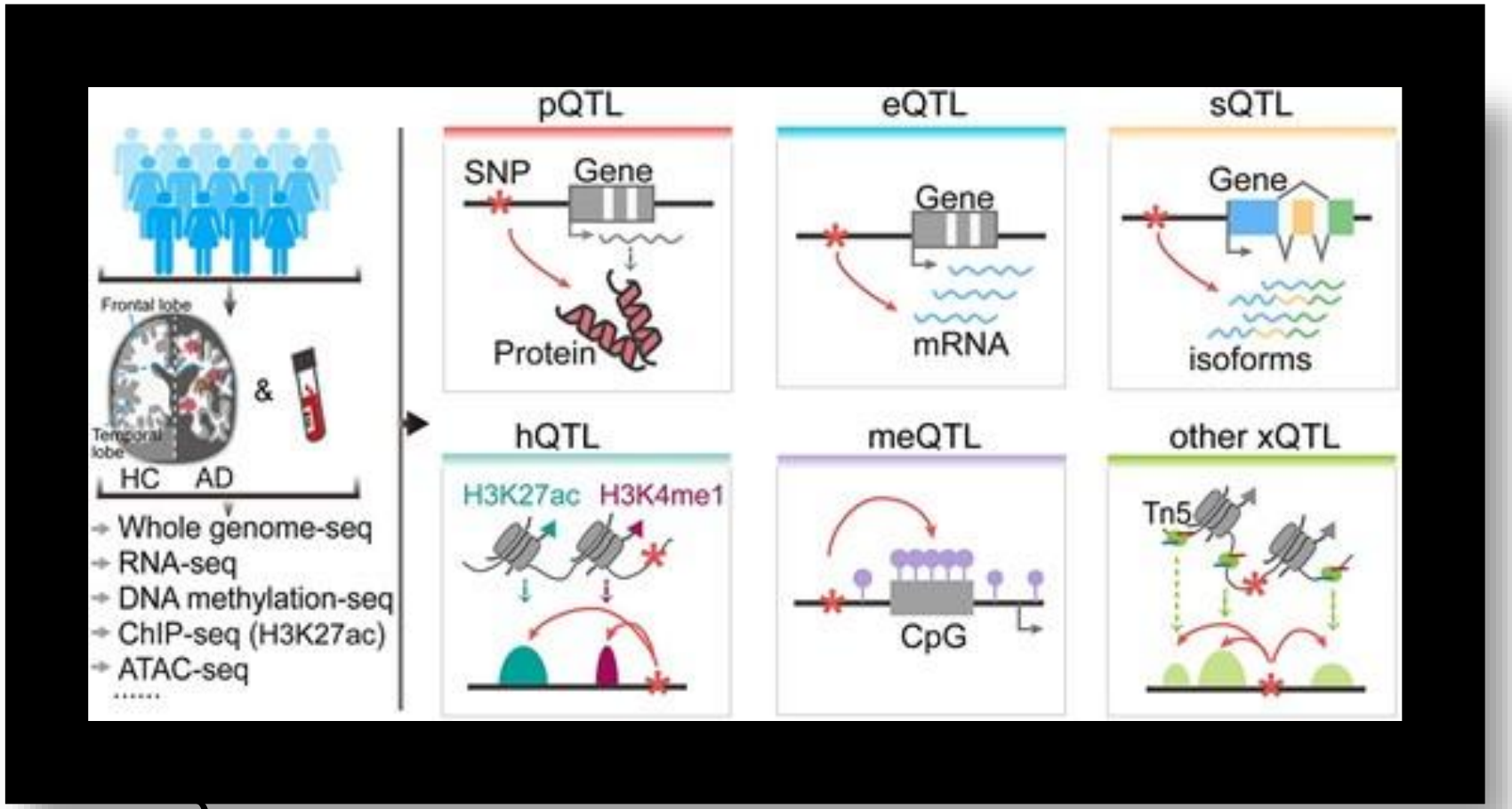
Genomika

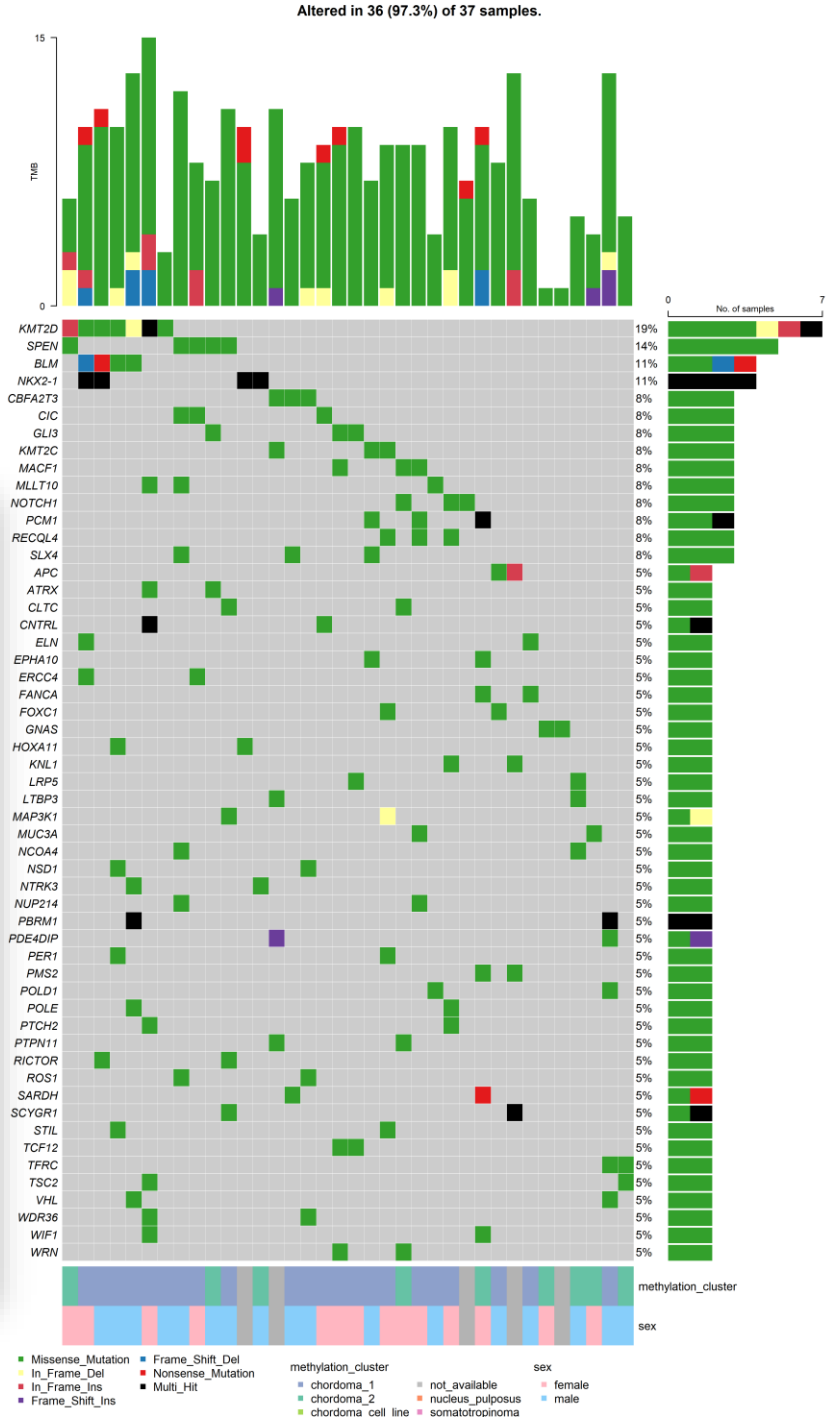

general transcription factors
RNA polymerase II
transcription regulators
coactivators
gene X
TATA
"spacer" DNA
cis-regulatory sequence
promoter
the gene control region for gene X


Mediator
gene X
TATA
RNA transcript


Regulatory sequence
Regulatory sequence
Enhancer /silencer
Promoter
5'UTR
Open reading frame
3'UTR
Enhancer /silencer
Proximal
Core
Start
Stop
Terminator
DNA
Transcription
Exon
Exon
Exon
Intron
Intron
Pre-mRNA
Post-transcription modification
Protein coding region
5'cap
Poly-A tail
Mature mRNA
Translation
Protein

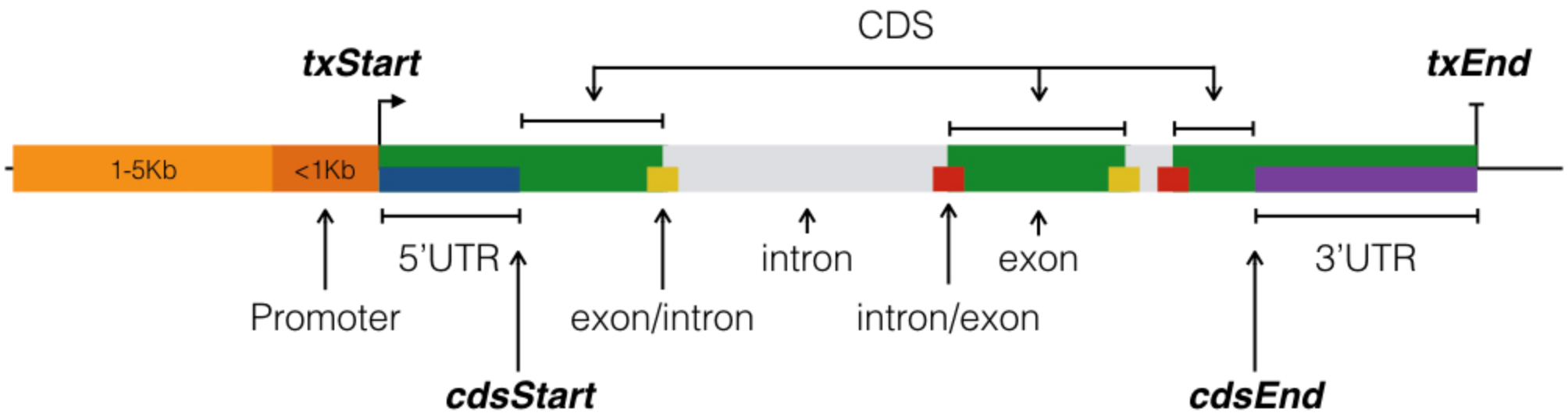
I co dalej?


Frontal lobe
Temporal lobe
HC
AD
&
Whole genome-seq
RNA-seq
DNA methylation-seq
ChIP-seq (H3K27ac)
ATAC-seq
pQTL
SNP
Gene
Protein
eQTL
Gene
mRNA
sQTL
Gene
isoforms
hQTL
H3K27ac
H3K4me1
meQTL
CpG
other xQTL
Tn5


Altered in 36 (97.3%) of 37 samples.
TMB
No. of samples
KMT2D 19%
SPEN 14%
BLM 11%
NKX2-1 11%
CBFA2T3 8%
CIC 8%
GLI3 8%
KMT2C 8%
MACF1 8%
MLLT10 8%
NOTCH1 8%
PCM1 8%
RECQL4 8%
SLX4 8%
APC 5%
ATRX 5%
CLTC 5%
CNTRL 5%
ELN 5%
EPHA10 5%
ERCC4 5%
FANCA 5%
FOXC1 5%
GNAS 5%
HOXA11 5%
KNL1 5%
LRP5 5%
LTBP3 5%
MAP3K1 5%
MUC3A 5%
NCOA4 5%
NSD1 5%
NTRK3 5%
NUP214 5%
PBRM1 5%
PDE4DIP 5%
PER1 5%
PMS2 5%
POLD1 5%
POLE 5%
PTCH2 5%
PTPN11 5%
RICTOR 5%
ROS1 5%
SARDH 5%
SCYGR1 5%
STIL 5%
TCF12 5%
TFRC 5%
TSC2 5%
VHL 5%
WDR36 5%
WIF1 5%
WRN 5%
methylation_cluster
sex
Missense_Mutation
In_Frame_Del
In_Frame_Ins
Frame_Shift_Ins
Frame_Shift_Del
Nonsense_Mutation
Multi_Hit
methylation_cluster
chordoma_1
chordoma_2
chordoma_cell_line
not_available
nucleus_pulposus
somatotropinoma
sex
female
male

Bazy danych

Genome browser: https://genome.ucsc.edu
Omim: https://www.omim.org
GeneBass: https://app.genebass.org



Normalizacja

https://hbctraining.github.io/DGE_workshop/lessons/02_DGE_count_normalization.html

Step	Description	TMM (edgeR)	RLE (DESeq2)	MRN
I	Pre-normalization by library size	$Y_{gkr} = \frac{X_{gkr}}{N_{kr}}$		$Y_{gkr} = \frac{X_{gkr}}{N_{kr}}$
II	Reference sample, or *pseudo-reference sample* (DESeq2)	$Y_g^{\mathrm{TMM}} = Y_{g11}$	$Y_g^{\mathrm{RLE}} = \sqrt[KR]{\prod_{k=1}^{K}\prod_{r=1}^{R} X_{gkr}}$	$Y_g^{\mathrm{MRN}} = \frac{1}{R}\sum_{r=1}^{R} Y_{g1r}$
III	Relative sizes of transcriptomes and reference sample, or *relative scaling factors* (edgeR), or *size factors* (DESeq2)	$\tau_{kr}^{\mathrm{TMM}} = \frac{1}{\#\mathcal{G}_{kr}^{*}}\sum_{g\in\mathcal{G}_{kr}^{*}}\frac{Y_{gkr}}{Y_g^{\mathrm{TMM}}}$ *where* $\mathcal{G}_{kr}^{*}$ represents the set of not trimmed genes	$\tau_{kr}^{\mathrm{RLE}} = \underset{g}{\mathrm{median}}\left(\frac{X_{gkr}}{Y_g^{\mathrm{LRE}}}\right)$	$\tau_k^{\mathrm{MRN}} = \underset{g}{\mathrm{median}}\left(\frac{\bar{Y}_{gk}}{Y_g^{\mathrm{MRN}}}\right)$ where $\bar{Y}_{gk} = \frac{1}{R}\sum_{r=1}^{R} Y_{gkr}$
IV	*Relative scaling factors* adjusted to multiply to 1 (edgeR)	$\tilde{\tau}_{kr}^{\mathrm{TMM}} = \frac{\tau_{kr}^{\mathrm{TMM}}}{\tilde{\tau}^{\mathrm{TMM}}}$ where $\tilde{\tau}^{\mathrm{TMM}} = \sqrt[KR]{\prod_{k=1}^{K}\prod_{r=1}^{R}\tau_{kr}^{\mathrm{TMM}}}$		
V	Taking into account both the relative size and the library size, or *effective library size* (edgeR)	$e_{kr}^{\mathrm{TMM}} = \tilde{\tau}_{kr}^{\mathrm{TMM}} N_{kr}$		$e_{kr}^{\mathrm{MRN}} = \tau_k^{\mathrm{MRN}} N_{kr}$
VI	Normalization factors, or *relative normalization factors* (edgeR), or *size factors* (DESeq2)	$f_{kr}^{\mathrm{TMM}} = \tilde{\tau}_{kr}^{\mathrm{TMM}}$	$f_{kr}^{\mathrm{RLE}} = \tau_{kr}^{\mathrm{RLE}}$	$f_{kr}^{\mathrm{MRN}} = \frac{e_{kr}^{\mathrm{MRN}}}{\tilde{e}^{\mathrm{MRN}}}$ where $\tilde{e}^{\mathrm{MRN}} = \sqrt[KR]{\prod_{k=1}^{K}\prod_{r=1}^{R} e_{kr}^{\mathrm{MRN}}}$
VII	Normalization of counts, or *counts-per-million* (edgeR)	$Z_{gkr}^{\mathrm{TMM}} = \frac{X_{gkr}}{e_{kr}^{\mathrm{TMM}}}10^6$	$Z_{gkr}^{\mathrm{RLE}} = \frac{X_{gkr}}{f_{kr}^{\mathrm{RLE}}}$	$Z_{gkr}^{\mathrm{MRN}} = \frac{X_{gkr}}{f_{kr}^{\mathrm{MRN}}}$

Jak uczyć się bioinformatyki?

1. ROBIĆ
2. ROBIĆ
3. ROBIĆ

BIOCONDUCTOR

RCOMPANION STATISTICS

RCOMPANION STATISTICS FOR BIOLOGICAL SCIENCE

R FOR DATA SCIENCE

STATQUEST

KOLEJNOŚĆ ALFABETYCZNA